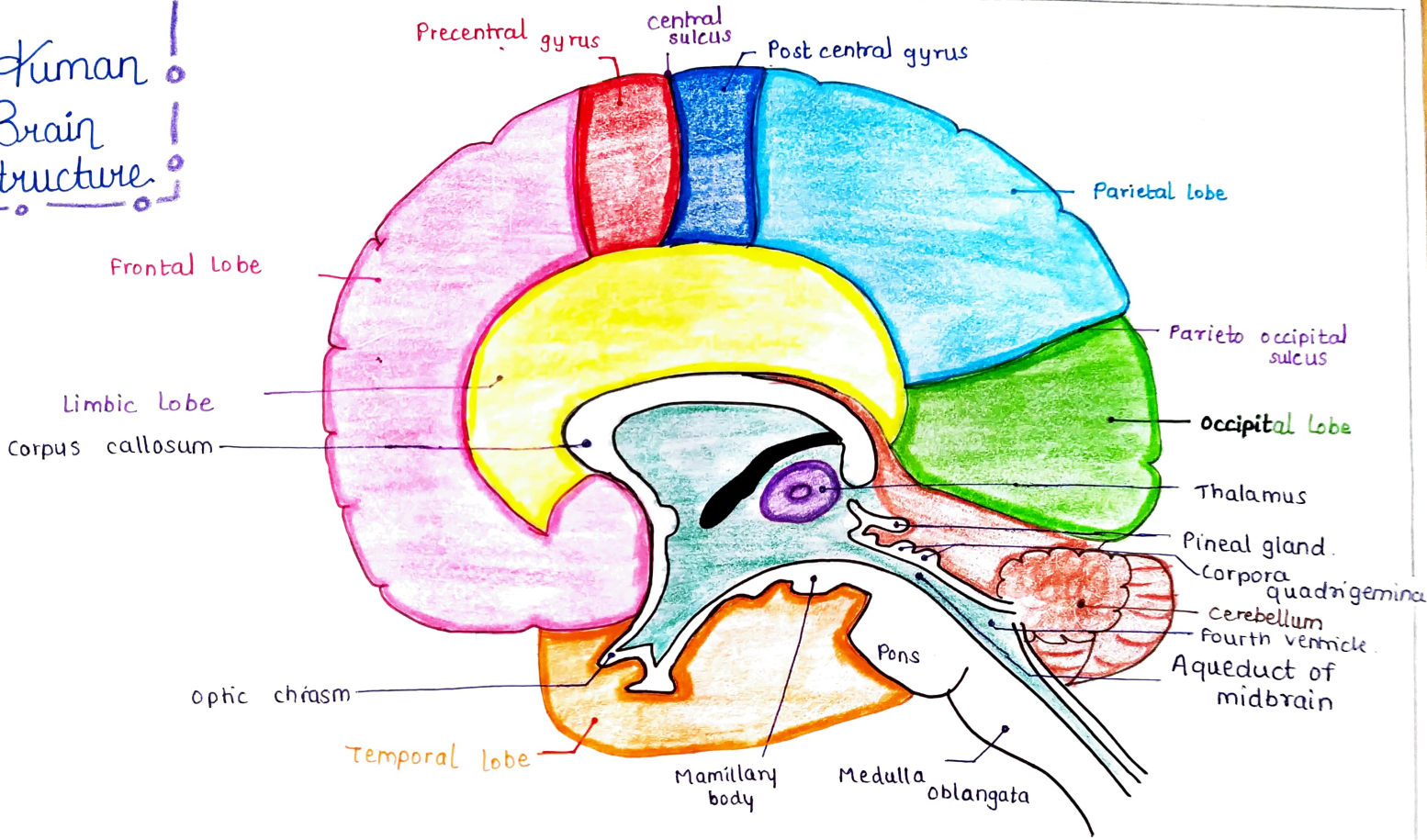
Human Brain Structure



Brain And its function.

1) The outer Shell: Lobes & Gyri

- These areas handle high-level processing and sensory awarness.

- **Frontal lobe**: executive unit or center, handles logic, planning, personality and motor control.

- **Precentral Gyrus**: Located in frontal lobe; it is the Primary Motor Cortex; responsible for sending signals tomove your muscles/our muscles.

- **Central Sulcus**: The physical "valley" or grooves that separates the frontal lobe from the parietal lobe.

- **Postcentral Gyrus**: Located in parietal lobe; it is the Primary Somatosensory cortex, which processes the sense of touch, pressure and pain.

- **Parietal lobe**: Integrates the sense of/sensory information and helps with spatial navigation (knowing where your body parts are).

- **Parieto-occipital sulcus**: The groove that marks the boundary betw parietal and occipital lobes.

Occipital lobes: The visual processing hub. It turns signals from your eyes into images.

- **Temporal lobe**: Responsible for hearing, language recognition and memory storage

2) Inner core: Coordination & communication.

- These structures sits deep within the brain to connect different regions.

1. **Corpus Callosum**: A thick bundle of nerve fibres that acts as abridge, allowing the left and right hemispheres to communicate.

2. **Limbic Lobe**:
- A ring-shaped region involved in your "drive" (emotions, behavior & long-term memory).

3. **Thalamus**:
- The Grand central station. it relays sensory and motor signals to cerebral cortex.

4. **Hypothalamus** (near Mammillary Body):
- controls the "Four Fs": Fighting, Fleeing, Feeding and mating.
- It regulates body temperature & hormones.

5. **Mammillary Body**
- Small round structures that are vital for recollective memory. (remembering specific events).

6. **Pineal Gland**:
- A small endocrine gland that produces melatonin, which regulates your sleep-wake cycle.

7. **Corpora Quadrigemina**:
- located in midbrain; these are reflex centers for vision (tracking moving objects) and hearing (turning your head towards a sound).

3) The fluid system & connections.

- The brain requires pathways for signals & fluid to flow.

1. Optic Chiasm:
The x-shaped structure where the optic nerves from eyes cross over to go to opposite sides of brain.

2. Aqueduct of Midbrain:
- A narrow canal that allows Cerebrospinal Fluid (CSF) to flow betw third & fourth ventricles.

3. Fourth ventricles:
- A diamond shaped cavity filled with CSF that protects the brain from trauma & helps circulate nutrients.

4) The Brainstem & Balance
These ~~primary~~ "primitive" areas keep you alive and upright.

1. Pons:
- Acts as a relay station betw the cerebrum & cerebellum; it also helps regulate breathing and sleep.

2. Medulla Oblongata:
- The most critical part for survival. It controls autonomic functions like your heartbeat, breathing and blood pressure.

3. Cerebellum:
- often called the "Little Brain". it ensures your movements are smooth, coordinated & balanced.

Some structures:

1) cerebrum (The 'thinking' Brain)
- Largest part ⇒ two hemispheres.
- Outer surface ⇒ cerebral cortex split into four functional lobes. (Frontal, Parietal, Temporal, Occipital)

2) Cerebellum ("The Little Brain")
- Located at back, under occipital lobe.
- Function: fine tunes provided along with handling balance, posture & coordination.

3) Brainstem ("The Life support")
- connects brain to spinal cord.
- Midbrain: controls eye moment & process auditory / visual relays.

4) Inner Brain
- Deep inside the cerebrum lies the limbic system, often "emotional brain".
- Thalamus: relay station.
- Hypothalamus: master controller of body's homeostasis.
  - regulates hunger, thirst, sleep & body temperature.
- Hippocampus: seahorse shaped, essential for forming the new long-term memories.
- Amygdala: Two almond-shaped structure ⇒ Fear & "Flight-fight" response controller.

Grey Matter	White Matter.
· Cell bodies & Dendrites	Myelinated Axons
· Processing & interpreting	Communication & Signaling
· Brain Location: Outer layer (Cortex)	Inner layer (Tracts)
· Spinal Location: Inner core (Butterfly shaped)	Outer layer